Assignment 1

Wireshark Packet Capture & Protocol Analysis

Objective

To capture live network packets using Wireshark and identify basic network protocols such as TCP, UDP, IP, and DNS.

Procedure

Step 1: Launch Wireshark

- Open Wireshark
- You will see a list of network interfaces:
  - Wi-Fi
  - Ethernet

Step 2: Start Packet Capture

- Double-click on Wi-Fi (or active interface)
- Packet capture will start immediately

Step 3: Generate Network Traffic

Perform any of the following:

- Open a web browser
- Visit any website (e.g., google.com)
- Open Command Prompt and run:
  - ping google.com

Step 4: Stop Capture

- Click the red Stop button
- Captured packets will remain on the screen

Step 5: Apply Basic Filters

Use the Display Filter bar:

Filter	Purpose
ip	Show IP packets
tcp	Show TCP packets
udp	Show UDP packets
dns	Show DNS packets
icmp	Show ping packets

Step 6: Analyze a Packet

- Click on any packet

- Observe:

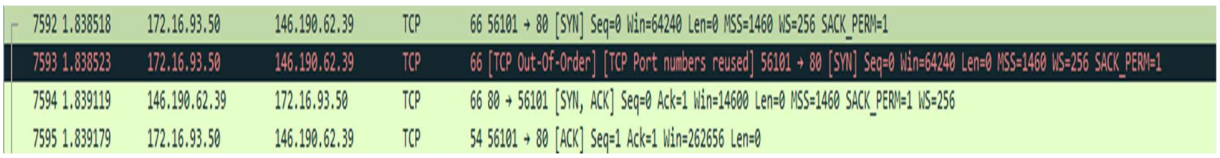
Header	Observed Value
Ethernet Header	Ethernet II, Src: HewlettP_22:9d:ca (f4:39:09:22:9d:ca), Dst: BrocadeC_90:80:78 (74:8e:f8:90:80:78)
IP Header	Internet Protocol Version 4, Src: 172.16.93.50, Dst: 23.34.253.33
TCP/UDP Header	Transmission Control Protocol, Src Port: 64114, Dst Port: 80, Seq: 1, Ack: 1, Len: 0
Payload Data	Application layer data (HTTP payload)

Observation

Parameter	Value
Interface used	Wi-Fi
Protocols observed	TCP
Source IP address	172.16.93.50
Destination IP address	23.34.253.33
Packet length	54 bytes

Result

Live network packets were successfully captured using Wireshark, and basic network protocols were identified and analysed.



7592	1.838518	172.16.93.50	146.190.62.39	TCP	66 56101 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
7593	1.838523	172.16.93.50	146.190.62.39	TCP	66 [TCP Out-Of-Order] [TCP Port numbers reused] 56101 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
7594	1.839119	146.190.62.39	172.16.93.50	TCP	66 80 → 56101 [SYN, ACK] Seq=0 Ack=1 Win=14600 Len=0 MSS=1460 SACK_PERM=1 WS=256
7595	1.839179	172.16.93.50	146.190.62.39	TCP	54 56101 → 80 [ACK] Seq=1 Ack=1 Win=262656 Len=0

```
C:\Users\gugnu>ipconfig

Windows IP Configuration


Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter WiFi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : fdc0:39c4:4c52:4d7a:d43f:4f1f:2448:3fbf
   Temporary IPv6 Address. . . . . . : fdc0:39c4:4c52:4d7a:111:421f:dfad:918d
   Link-local IPv6 Address . . . . . : fe80::9a41:687c:51b3:a52f%6
   IPv4 Address. . . . . . . . . . . : 172.17.67.150
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . : 172.17.64.1

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
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C:\Users\gugnu>ping 172.17.64.1

Pinging 172.17.64.1 with 32 bytes of data:
Reply from 172.17.64.1: bytes=32 time=4ms TTL=128
Request timed out.
Reply from 172.17.64.1: bytes=32 time=3ms TTL=128
Reply from 172.17.64.1: bytes=32 time=2ms TTL=128

Ping statistics for 172.17.64.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 4ms, Average = 3ms
```